Exact polynomial moment boundaries
Gaussian–cubic truncations and pure-order positivity

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Reviewed editorial draft, October 30, 2026

Abstract

We determine positivity of polynomial squares through degree four and realization by positive measures through degree nine for the formal moment sequence with exponential generating series $\exp(vz^2/2 + kz^3/6)$. For $v > 0$, put $r = k^2/v^3$. Quadratic, cubic and quartic square tests give successively

$$r \le 2, \qquad 12 - 24r - 9r^2 \ge 0, \qquad 8 - 40r - 45r^3 \ge 0.$$

The last inequality is exactly the criterion for matching moments through degree eight by a positive probability measure. Matching the prescribed ninth moment requires the strict inequality. At the quartic null boundary a positive measure matches all eight moments, but the null polynomial forces a different ninth moment. A parallel obstruction occurs between degrees six and seven. Finite Jacobi matrices construct the representing measures and steer an arbitrary ninth moment in the strict region. A sharp attained sixth-moment floor for the entire regular variance/skewness family gives a concrete low-degree diagnostic. A separate singular positive cubic-square functional has no measure representation but is a limit of genuine moment vectors, with an explicit escaping sixth-moment defect. For the separate pure-order generating series e^{az^p}, we give the complete all-degree square-positivity classification and an explicit negative-square witness. The methods are classical truncated moment theory; the explicit parameter formulas are matched to a frozen Lean development, with no priority claim.

1 The polynomial functional and the main boundaries

Let $v, k \in \mathbb{R}$. Define $m_0 = 1$ and, for $n \ge 0$,

$$m_{n+1} = vn\, m_{n-1} + \frac{k}{2} n(n-1) m_{n-2}, \tag{1}$$

where terms with negative indices are zero. Define the linear functional $\mathcal{L}_{v,k}$ on $\mathbb{R}[x]$ by $\mathcal{L}_{v,k}(x^n) = m_n$. The generating series identity

$$\sum_{n \ge 0} m_n \frac{z^n}{n!} = \exp(vz^2/2 + kz^3/6) \tag{2}$$

is an identity of formal power series: differentiation gives (1) and the constant coefficient is one. It is not the assumption that a positive law has this moment-generating function.

Say that $\mathcal{L}_{v,k}$ passes the degree-d square test when $\mathcal{L}_{v,k}(p^2) \ge 0$ for every real polynomial of degree at most d. A Borel probability measure represents the sequence through degree N when $\int |x|^j \, d\mu < \infty$ and $\int x^j \, d\mu = m_j$ for $0 \le j \le N$. Introduce

$$\begin{aligned} \delta &= 2v^3 - k^2, \\ D &= 12v^6 - 24v^3k^2 - 9k^4, \\ Q &= 288v^{10} - 1440v^7k^2 - 1620vk^6. \end{aligned} \tag{3}$$

Theorem 1.1. *For $v > 0$ the degree-two, degree-three and degree-four square tests hold exactly when $\delta \geq 0$, $D \geq 0$ and $Q \geq 0$, respectively. A positive probability measure represents the moments through degree eight exactly when $Q \geq 0$, and through degree nine exactly when $Q > 0$. All positive constructions may be chosen with finite support. If $Q > 0$, any real ninth moment can be prescribed while keeping the first eight moments fixed.*

The case $v = 0$ is degenerate: a positive measure with the first three prescribed moments must be δ_0 and $k = 0$. If $v < 0$, its second moment already excludes a positive measure. Thus the full degree-eight criterion is $(v, k) = (0, 0)$ or $v > 0, Q \geq 0$; at degree nine replace $Q \geq 0$ by $Q > 0$.

The issue at $Q = 0$ is not failure of the preceding square test. It is compatibility of the next moment with the null polynomial. We give the positive constructions and the incompatibility proof explicitly.

The general tools are classical. Curto and Fialkow's odd truncated Hamburger criterion tests both positivity and a range condition; their even criterion likewise distinguishes singular data [1]. Here the contribution under examination is an explicit evaluation for a two-parameter sequence and a verified exposition of its boundaries. We do not claim a new general moment theorem or established novelty for these formulas.

2 Moment matrices and exact square tests

The recursion gives

j	m_j	j	m_j
0	1	5	$10vk$
1	0	6	$15v^3 + 10k^2$
2	v	7	$105v^2k$
3	k	8	$105v^4 + 280vk^2$
4	$3v^2$	9	$1260v^3k + 280k^3$

(4)

Let $H_d = (m_{i+j})_{0 \leq i,j \leq d}$. The square test is exactly $H_d \succeq 0$, since the coefficient vector of p has quadratic form $\mathcal{L}_{v,k}(p^2)$.

Use the monic centered basis

$$U_0 = 1, \qquad U_1 = x, \qquad U_2 = x^2 - v,$$
$$U_3 = x^3 - 3vx - k, \qquad U_4 = x^4 - 6vx^2 - 4kx + 3v^2.$$

Each U_j, $j \geq 1$, has zero functional mean. This change of basis is triangular with diagonal one. On (U_1, U_2, U_3) the Gram matrix is

$$G_3 = \begin{pmatrix} v & k & 0 \\ k & 2v^2 & 6vk \\ 0 & 6vk & 6v^3 + 9k^2 \end{pmatrix}, \qquad \det G_3 = D. \tag{5}$$

Adjoining U_4 gives

$$G_4 = \begin{pmatrix} v & k & 0 & 0 \\ k & 2v^2 & 6vk & 6k^2 \\ 0 & 6vk & 6v^3 + 9k^2 & 36v^2k \\ 0 & 6k^2 & 36v^2k & 24v^4 + 144vk^2 \end{pmatrix}, \qquad \det G_4 = Q. \tag{6}$$

These identities follow by multiplying the displayed polynomials and using (4). In particular the leading determinants of H_2, H_3, H_4 are δ, D, Q.

Proposition 2.1. *For $v > 0$ the three square-test criteria in Theorem 1.1 hold.*

Proof. For $p = c + bx + ax^2$, direct completion of squares yields

$$v\mathcal{L}_{v,k}(p^2) = v(c + va)^2 + (vb + ka)^2 + \delta a^2.$$

This proves sufficiency of $\delta \geq 0$. If $\delta < 0$, take $(c, b, a) = (-v^2, -k, v)$; its square has value $v\delta < 0$.

Necessity of $D \geq 0$ and $Q \geq 0$ follows because a positive semidefinite matrix has nonnegative determinant. For sufficiency put $r = k^2/v^3 \geq 0$. If $D \geq 0$, then $12 - 24r - 9r^2 \geq 0$, which forces $r < 1/2$ and therefore $\delta > 0$. The leading 2×2 block of G_3 is positive definite. Its scalar Schur complement is $D/\delta \geq 0$, so $G_3 \succeq 0$. If $Q \geq 0$, its factorization

$$Q = 36v^{10}(8 - 40r - 45r^3) \tag{7}$$

implies $r \leq 1/5$. Hence $D/v^6 \geq 12 - 24/5 - 9/25 = 171/25 > 0$. Thus G_3 is positive definite and the last Schur complement in G_4 is $Q/D \geq 0$. This proves $G_4 \succeq 0$. The constant basis vector is orthogonal to these centered blocks and has norm one, completing the argument for H_d. □

This proof also identifies all positive pivots needed in the representing matrices below. To display exact null directions, define

$$R_3 = 6vk^2U_1 - 6v^2kU_2 + \delta U_3, \tag{8}$$

$$\begin{aligned} R_4 = {} & -18k^3(10v^3 - 3k^2)U_1 + 18vk^2(10v^3 - 3k^2)U_2 \\ & - 72v^2k(v^3 - k^2)U_3 + DU_4. \end{aligned} \tag{9}$$

Multiplication by the Gram matrices shows that R_3 is orthogonal to every polynomial of degree at most two, and R_4 to every polynomial of degree at most three. Their square values are

$$\mathcal{L}_{v,k}(R_3^2) = \delta D, \qquad \mathcal{L}_{v,k}(R_4^2) = DQ. \tag{10}$$

When the preceding pivots are positive, the monic orthogonal polynomials have squared norms

$$h_0 = 1, \qquad h_1 = v, \qquad h_2 = \delta/v, \qquad h_3 = D/\delta, \qquad h_4 = Q/D. \tag{11}$$

For example, $P_2 = U_2 - (k/v)U_1$, $P_3 = R_3/\delta$ and $P_4 = R_4/D$. These computations can be checked from the small Gram matrices without expanding large determinants.

3 A finite spectral construction, including a null last pivot

We spell out the finite construction to handle the semidefinite boundary rather than invoke positivity alone. The lemma is standard Jacobi-matrix moment theory.

Lemma 3.1. *Let L be a real linear functional on polynomials of degree at most $2d$, with $L(1) = 1$. Suppose H_{d-1} is positive definite and H_d is positive semidefinite. There is a real symmetric $(d + 1) \times (d + 1)$ matrix J_f, depending on one real parameter f, for which*

$$\langle e_0, J_f^n e_0 \rangle = L(x^n), \qquad 0 \leq n \leq 2d.$$

Its spectral probability measure at e_0 represents these moments. If h_d is the last monic orthogonal squared norm, then

$$\langle e_0, J_f^{2d+1} e_0 \rangle = \langle e_0, J_0^{2d+1} e_0 \rangle + h_d f. \tag{12}$$

The construction remains valid when $h_d = 0$.

Proof. Give $V_{d-1} = \mathbb{R}[x]_{\le d-1}$ the positive definite inner product $L(pq)$. Let $q_0 = 1, q_1, \dots, q_{d-1}$ be its orthonormal polynomials with positive leading coefficients. Orthogonal projection of x^d onto V_{d-1} defines the monic residual P_d and $h_d = L(P_d^2) \ge 0$. For $p = p_{<d} + cx^d$ in V_d, define a coordinate map into $\mathbb{R}^{d+1}$ by taking the coordinates of $p - cP_d \in V_{d-1}$ and giving it last coordinate $c\sqrt{h_d}$. Orthogonality proves

$$\langle C(p), C(q)\rangle = L(pq), \qquad p, q \in V_d. \tag{13}$$

When $h_d = 0$ this is a map of the semidefinite quotient; no division by h_d is used.

Specify the first d columns of J_f by $J_f C(p) = C(xp)$ for $p \in V_{d-1}$. These columns are symmetric on their first d entries, because $L(xpq) = L(pxq)$. Fill the remaining off-diagonal entries by symmetry, and set the last diagonal to f. This defines a real symmetric matrix, including at $h_d = 0$. By induction $J_f^j e_0 = C(x^j)$ for $0 \le j \le d$: only the first d columns are used until the last step. For $n \le 2d$ choose $i, j \le d$ with $i + j = n$. Then

$$\langle e_0, J_f^n e_0\rangle = \langle J_f^i e_0, J_f^j e_0\rangle = L(x^{i+j})$$

by (13).

Let $J_f = \sum_\lambda \lambda \Pi_\lambda$ be its finite orthogonal spectral decomposition. The measure

$$\mu_f = \sum_\lambda \|\Pi_\lambda e_0\|^2 \delta_\lambda \tag{14}$$

has nonnegative weights summing to one. Its nth moment is exactly $\langle e_0, J_f^n e_0\rangle$, and finite support supplies all absolute moments.

Finally $J_f - J_0 = f e_d e_d^{\mathsf{T}}$, while $J_f^d e_0 = C(x^d)$ is independent of f and has last coordinate $\sqrt{h_d}$. Thus

$$\langle e_0, J_f^{2d+1} e_0\rangle = \langle C(x^d), J_f C(x^d)\rangle = \langle C(x^d), J_0 C(x^d)\rangle + h_d f.$$

This proves (12). □

The symmetric matrices in the lemma are tridiagonal. Indeed if $i < j - 1$, orthogonality gives $L(xq_i q_j) = 0$, because xq_i has degree below j. If P_j is monic with norm $h_j > 0$, the neighboring entry is $\sqrt{h_j/h_{j-1}}$ by its leading coefficient. At $h_d = 0$, the last row and column decouple. The spectral measure at e_0 then lives on the preceding block; an arbitrary last diagonal introduces no mass.

4 Eight moments and a freely chosen ninth moment

Assume $v > 0$ and $Q \ge 0$. By Proposition 2.1, H_3 is positive definite and H_4 positive semidefinite. Lemma 3.1 with $d = 4$ gives a positive probability measure representing all moments through eight. Conversely such a measure makes every degree-four square nonnegative, so $Q \ge 0$. This proves the degree-eight classification, including its boundary.

For direct reproduction, the matrix is

$$J_f = \begin{pmatrix} 0 & b_1 & 0 & 0 & 0 \\ b_1 & a_1 & b_2 & 0 & 0 \\ 0 & b_2 & a_2 & b_3 & 0 \\ 0 & 0 & b_3 & a_3 & b_4 \\ 0 & 0 & 0 & b_4 & f \end{pmatrix}, \tag{15}$$

where

$$b_1 = \sqrt{v}, \qquad b_2 = \sqrt{\delta}/v, \qquad b_3 = \sqrt{vD}/\delta, \qquad b_4 = \sqrt{\delta Q}/D, \tag{16}$$

$$a_1 = k/v, \qquad a_2 = \frac{k(4v^3 + k^2)}{v\delta}, \qquad a_3 = \frac{72v^2k(v^3 - k^2)}{D} - \frac{6v^2k}{\delta}. \tag{17}$$

All denominators are strictly positive. The off-diagonal entries follow from (11); the diagonal entries are $\mathcal{L}_{v,k}(xP_j^2)/h_j$ for $j = 1, 2, 3$. For example $\mathcal{L}_{v,k}(xP_1^2) = k$, and inserting $P_2 = x^2 - (k/v)x - v$ gives $\mathcal{L}_{v,k}(xP_2^2) = 4vk + k^3/v^2$, which yields a_2. The same substitution of R_3/δ gives a_3. Thus (15) is the concrete form of the construction, not a matrix whose moment matching is assumed.

Write $s_9(f) = \int x^9 \, d\mu_f$. The steering formula is

$$s_9(f) = s_9(0) + (Q/D)f. \tag{18}$$

If $Q > 0$, any target $s \in \mathbb{R}$ is obtained by $f = (s - s_9(0))D/Q$. In particular one can choose $s = m_9 = 1260v^3k + 280k^3$. This proves the positive degree-nine construction and the arbitrary-target assertion. At $Q = 0$, however, (18) has zero slope. The next section shows that the prescribed m_9 is incompatible with every positive representation, not merely with this chosen family of matrices.

5 The null boundary excludes the prescribed ninth moment

We first record the general compatibility argument.

Lemma 5.1. *Suppose a positive measure represents L through degree $2d + 1$, and p has degree at most d with $L(p^2) = 0$. Then $L(xp^2) = 0$.*

Proof. Polynomial integrability follows from the assumed absolute moments. Since $p^2 \geq 0$ and its integral is zero, $p = 0$ almost everywhere. Therefore $xp^2 = 0$ almost everywhere, and its integral is zero. □

For the particular polynomial R_4 of (9), direct multiplication with the moments through degree nine gives the exact identity

$$\begin{aligned} v\mathcal{L}_{v,k}(xR_4^2) = {} & (-9k^5 + 48v^3k^3 - 108v^6k)Q \\ & + 864v^{10}k(52v^6 - 272v^3k^2 + 213k^4). \end{aligned} \tag{19}$$

This is a polynomial identity in v, k, verifiable from the finite table; the accompanying symbolic checker independently verifies its transcription.

Now assume $v > 0, Q = 0$. Its normalized equation is $8 - 40r - 45r^3 = 0$, where $r = k^2/v^3 \geq 0$. It implies $r \leq 1/5$, and $k \neq 0$ since the constant term is eight. On $0 \leq r \leq 1/5$ the polynomial $52 - 272r + 213r^2$ is decreasing, as its derivative $-272 + 426r$ is negative. Its minimum on this interval is $153/25 > 0$. Hence the second line of (19) is nonzero, and $\mathcal{L}_{v,k}(xR_4^2) \neq 0$. But (10) gives $\mathcal{L}_{v,k}(R_4^2) = 0$. Lemma 5.1 rules out a positive degree-nine representation. Together with the preceding section, this completes Theorem 1.1 for $v > 0$.

The degenerate cases stated after the theorem also follow without a moment-extension theorem. If $v = 0$, a positive representing measure has $x = 0$ almost everywhere, so $k = 0$ and it must be δ_0. Conversely (1) gives $m_j = 0$ for every $j > 0$ when $(v, k) = (0, 0)$. Negative v is impossible because $\int x^2 \, d\mu \geq 0$.

The boundary is nonempty. The polynomial $g(r) = 8 - 40r - 45r^3$ is strictly decreasing for $r \geq 0$, with $g(0) > 0$ and $g(1/5) < 0$. It has a unique positive root r_4, and $g(1/6) = 9/8 > 0$ shows

$$1/6 < r_4 < 1/5. \tag{20}$$

For $v = 1$, $k = \sqrt{r_4}$ supplies an explicit algebraic parameter at which all eight moments are realizable but the prescribed ninth is not.

6 The preceding six-to-seven obstruction

The smaller null boundary illustrates the same mechanism at a different truncation and is part of the same story.

Proposition 6.1. *If $v > 0$ and $D = 0$, a positive probability measure represents the moments through degree six, but none represents them through degree seven.*

Proof. The equation $D = 0$ gives $12 - 24r - 9r^2 = 0$, with

$$r = r_3 = (2\sqrt{7} - 4)/3 < 1/2. \tag{21}$$

Thus $\delta > 0$, H_2 is positive definite and H_3 is positive semidefinite. Lemma 3.1 with $d = 3$ constructs the degree-six representation. The last pivot $h_3 = D/\delta$ is zero, so it suffices to retain the preceding three-dimensional block.

The null cubic R_3 satisfies $\mathcal{L}_{v,k}(R_3^2) = \delta D = 0$. Its next cross moment is

$$\mathcal{L}_{v,k}(xR_3^2) = -14v^2kD + 24v^5k(10v^3 - 17k^2). \tag{22}$$

Again this follows by polynomial multiplication and (4). At $D = 0$ one has $k \neq 0$ and $10v^3 - 17k^2 = v^3(10 - 17r_3) > 3v^3/2 > 0$. So (22) is nonzero, contradicting Lemma 5.1 for any degree-seven representation. □

There is also an open gap between the cubic and quartic square tests. At $v = 1, k = 1/2$, one has $r = 1/4$ and

$$D = 87/16 > 0, \qquad Q/36 = 8 - 40/4 - 45/64 = -173/64 < 0.$$

Continuity preserves both strict signs on a neighborhood. Thus even strict cubic square positivity does not guarantee quartic square positivity or eight-moment realizability. This example is a direct specialization of the exact criteria, not an appeal to numerical eigenvalues.

7 An attained sixth-moment bound

The regular variance/skewness family admits an exact sixth-moment floor at every prescribed fifth moment, not only at $v = k = 1$. The construction is a concrete instance of the classical finite Jacobi spectral argument already used above.

Proposition 7.1. *Fix $v > 0$, $k, t \in \mathbb{R}$ and $\delta = 2v^3 - k^2 > 0$. If a probability measure has finite sixth absolute moment and*

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = v, \quad m_3 = k, \quad m_4 = 3v^2, \quad m_5 = t,$$

then

$$m_6 \geq F(v,k,t) := 9v^3 + k^2 + \frac{v(t-4vk)^2}{\delta}. \tag{23}$$

For every such v, k, t, equality is attained by a probability measure with at most three support points and all absolute moments finite.

Proof. Set $u = t - 4vk$ and define

$$\alpha = \frac{vu}{\delta}, \qquad \beta = 3v - \frac{ku}{\delta}, \qquad \gamma = k - \frac{v^2 u}{\delta}, \qquad q(x) = x^3 - \alpha x^2 - \beta x - \gamma.$$

The given moments imply the three exact orthogonality identities

$$\int q\, d\mu = k - \alpha v - \gamma = 0, \qquad \int xq\, d\mu = 3v^2 - \alpha k - \beta v = 0,$$

$$\int x^2 q\, d\mu = t - 3\alpha v^2 - \beta k - \gamma v = 0.$$

For the last equality, substitution reduces its subtracted part to $4vk + (2v^3 - k^2)u/\delta = 4vk + u = t$. Thus q is orthogonal to $r(x) = \alpha x^2 + \beta x + \gamma$. Since $q = x^3 - r$,

$$\int q^2\, d\mu = m_6 - (\alpha t + 3\beta v^2 + \gamma k) = m_6 - F(v,k,t).$$

Indeed the parenthesis equals $9v^3 + k^2 + vu(t-4vk)/\delta$. Nonnegativity of the actual square integral proves (23). The same proof works for any square-positive polynomial functional with these moments.

For attainment let

$$d = \alpha - \frac{k}{v} = \frac{v^2 t - 6v^3 k + k^3}{v\delta}, \qquad J = \begin{pmatrix} 0 & \sqrt{v} & 0 \\ \sqrt{v} & k/v & \sqrt{\delta}/v \\ 0 & \sqrt{\delta}/v & d \end{pmatrix}.$$

This is a real symmetric matrix for every allowed parameter. If $J = \sum_\lambda \lambda P_\lambda$ is its orthogonal spectral decomposition, define $\mu_J = \sum_\lambda \|P_\lambda e_0\|^2 \delta_\lambda$. Its nonnegative weights sum to $\|e_0\|^2 = 1$, and its at most three atoms give all absolute moments. For nonnegative integers r, s,

$$\int x^{r+s}\, d\mu_J = \langle J^r e_0, J^s e_0 \rangle.$$

The necessary coordinate vectors can be computed without a large matrix-power expansion:

$$Je_0 = (0, \sqrt{v}, 0), \qquad J^2 e_0 = \left(v, \frac{k}{\sqrt{v}}, \frac{\sqrt{\delta}}{\sqrt{v}}\right),$$

$$J^3 e_0 = \left(k, 3v\sqrt{v}, \frac{\alpha\sqrt{\delta}}{\sqrt{v}}\right).$$

For the second entry of the last vector use $k^2 + \delta = 2v^3$; for the third use $d + k/v = \alpha$. These vectors give $m_0 = 1, m_1 = 0, m_2 = v, m_3 = k$ and

$$m_4 = v^2 + \frac{k^2 + \delta}{v} = 3v^2, \qquad m_5 = 4vk + \frac{\alpha\delta}{v} = t,$$

$$m_6 = k^2 + 9v^3 + \frac{\alpha^2 \delta}{v} = F(v,k,t).$$

Thus the claimed bound is attained in an actual probability law. Its integral of q^2 is zero, so $q = 0$ almost everywhere under that law and its support is contained in the real zero set of q. No moment beyond those displayed is prescribed by this construction. □

At $v = k = 1$ one has $\delta = 1$ and recovers

$$m_6 \geq t^2 - 8t + 26. \tag{24}$$

Here $q = p_t = x^3 + (4-t)x^2 + (t-7)x + (t-5)$ and the same spectral matrix becomes $\left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & t-5 \end{smallmatrix}\right)$. For the mixed Gaussian–cubic sequence, substitute $t = 10vk$ to obtain

$$15v^3 + 10k^2 - F(v, k, 10vk) = \frac{12v^6 - 24v^3k^2 - 9k^4}{\delta} = \frac{D}{\delta}.$$

Thus the earlier cubic-square boundary has a sharp attained sixth-moment interpretation throughout the regular region. The singular case $\delta = 0$ is excluded from this division and is not settled by a limiting convention.

For the prescribed sequence at $v = k = 1$, $m_5 = 10$ and $m_6 = 25$, whereas (24) requires $m_6 \geq 46$. Indeed $p_{10} = x^3 - 6x^2 + 3x + 5$ satisfies $\mathcal{L}_{1,1}(p_{10}^2) = -21$. The degree-two square test still passes, since $k^2 = 1 \leq 2v^3$. This contrasts a finite positive truncation with the next genuinely incompatible data.

8 A distinct singular even truncation and escaping sixth moment

The preceding Gaussian–cubic boundaries have positive preceding Gram blocks, allowing the specified finite Jacobi construction. Singular even data without that hypothesis can be positive on every available square and still have no representing measure. We give an actual convergent probability sequence illustrating precisely that obstruction. This is a separate moment vector, not another parameter value in $\exp(vz^2/2 + kz^3/6)$.

8.1 Positive cubic squares with an incompatible null relation

Let $\Lambda(x^j) = s_j$ for $0 \leq j \leq 6$, where

$$(s_0, s_1, s_2, s_3, s_4, s_5, s_6) = (1, 0, 2, 4, 12, 32, 89), \qquad R(x) = x^2 - 2x - 2.$$

Every polynomial g of degree at most three has a unique expansion $g = c + bx + aR + dxR$, since these four polynomials have successive degrees zero, one, two and three and leading coefficients one. The exact Gram matrix of this basis for $\Lambda(fg)$ is

$$\operatorname{diag}(1, 2, 0, 1). \tag{25}$$

Here is a direct verification of all off-diagonal cancellations:

$$\Lambda(x) = \Lambda(R) = \Lambda(xR) = \Lambda(x^2R) = \Lambda(xR^2) = 0.$$

For example $\Lambda(x^2R) = 12 - 2 \cdot 4 - 2 \cdot 2 = 0$, and $\Lambda(xR^2) = 32 - 4 \cdot 12 + 8 \cdot 2 + 4 \cdot 0 = 0$. Also

$$R^2 = x^4 - 4x^3 + 8x + 4, \qquad (xR)^2 = x^6 - 4x^5 + 8x^3 + 4x^2,$$

so $\Lambda(R^2) = 12 - 16 + 4 = 0$ and $\Lambda((xR)^2) = 89 - 128 + 32 + 8 = 1$. Together with $\Lambda(1) = 1, \Lambda(x^2) = 2$, these are every entry of (25). Consequently

$$\Lambda(g^2) = c^2 + 2b^2 + d^2 \geq 0 \quad (\deg g \leq 3).$$

Nevertheless no positive probability measure has these seven moments. Such a measure would give $\int R^2 = 0$, forcing $R = 0$ almost everywhere by nonnegativity. Hence $xR = 0$ almost everywhere, contradicting $\int (xR)^2 = 1$. All these polynomials are integrable because the requested sixth absolute moment is finite. Thus positivity of a singular cubic-square test is not alone the even truncated realization theorem.

There is also no normed coordinate-multiplication model preserving these data on the stated degrees. If a linear map U from polynomials into a normed space satisfied $\|Ug\|^2 = \Lambda(g^2)$ for $\deg g \le 3$, then $UR = 0$. If a linear operator T also satisfied $U(xg) = T(Ug)$ for $\deg g \le 2$, it would force $U(xR) = 0$, whereas its squared norm is one. The moment Gram quotient exists as a finite semidefinite quotient; it is the proposed coordinate action on that quotient that fails to be well-defined.

8.2 An explicit two-atom law and a vanishing remote atom

Set $\lambda_\pm = 1 \pm \sqrt{3}$ and define

$$\nu = \frac{\sqrt{3}-1}{2\sqrt{3}}\delta_{\lambda_+} + \frac{\sqrt{3}+1}{2\sqrt{3}}\delta_{\lambda_-}.$$

Its weights are positive, sum to one and give mean zero. Both support points solve $x^2 = 2x + 2$, so its moments b_j satisfy $b_0 = 1, b_1 = 0$ and $b_j = 2b_{j-1} + 2b_{j-2}$. Thus

$$(b_0, b_1, b_2, b_3, b_4, b_5, b_6) = (1, 0, 2, 4, 12, 32, 88).$$

This is also the coordinate-zero spectral measure of $\left(\begin{smallmatrix} 0 & \sqrt{2} \\ \sqrt{2} & 2 \end{smallmatrix}\right)$: its two distinct eigenvalues are $\lambda_\pm$, and its unit spectral weights have mean the zero upper-left entry, determining the same displayed weights. This identifies the human two-atom formula with the formal source's actual spectral law.

For integers $N \ge 1$ put

$$\nu_N = (1 - N^{-6})\nu + N^{-6}\delta_N.$$

Every ν_N is a positive probability with finite support and all absolute moments. For $0 \le j \le 6$ the exact formula is

$$\int x^j \, d\nu_N = (1 - N^{-6})b_j + N^{j-6}. \tag{26}$$

For $j < 6$ it tends to $b_j = s_j$, whereas for $j = 6$ it tends to $88 + 1 = 89 = s_6$. Thus genuine probability moment vectors converge in $\mathbb{R}^7$ to the nonrepresentable vector just proved positive on cubic squares. The set of realizable probability moment vectors through degree six is therefore not closed. This is nonclosedness of the actual unbounded-support moment image, not a failure of closedness of the positive-semidefinite matrix cone containing it.

For every bounded Borel f the actual law difference obeys

$$\left| \int f \, d\nu_N - \int f \, d\nu \right| = N^{-6} \left| f(N) - \int f \, d\nu \right| \le 2N^{-6}\|f\|_\infty.$$

In particular $\nu_N \to \nu$ weakly, indeed in the total-variation norm bounded by $2N^{-6}$. This quantitative total-variation bound is a printed elementary deduction; the mapped source directly gives convergence on every bounded observable. The weak limit's sixth moment is 88, not the limiting moment coordinate 89. No alternative weak limit represents the displayed limiting vector.

8.3 Tightness and a bounded sixth moment are insufficient

These laws even have the uniform bound

$$\int |x|^6 \, d\nu_N = 89 - 88N^{-6} \le 89.$$

Markov's elementary bound gives $\nu_N\{|x| > R\} \le 89/R^6$ for $R > 0$, hence uniform tightness. But for every $R > 0$, choose $N > R$. The remote atom alone then contributes

$$\int_{|x|>R} |x|^6 \, d\nu_N \ge N^{-6} N^6 = 1.$$

Thus the sixth powers are not uniformly integrable under these laws. The atom's probability mass vanishes while its sixth-moment contribution stays exactly one. This is the missing hypothesis in any inference from the weak law to the highest moment. For comparison, if $0 \le q < 6$, then on $|x| > R$, $|x|^q \le R^{q-6}|x|^6$, so the corresponding tails are uniformly bounded by $89R^{q-6} \to 0$. This lower-power estimate is an explicit classical deduction, consistent with the convergent lower coordinates in (26). The construction illustrates standard noncompact moment geometry and uniform-integrability limits; no new general nonclosedness or weak-convergence principle is claimed.

9 The separate pure-order Gaussian boundary

The mixed Gaussian–cubic truncation is different from a pure-order functional. To make the distinction exact, fix an integer $p \ge 2$ and $a \in \mathbb{R}$. Define $L_{p,a}$ by the formal coefficient identity

$$L_{p,a}(e^{xz}) = e^{az^p}, \qquad L_{p,a}(x^n) = \begin{cases} (pj)!a^j/j!, & n = pj, \\ 0, & p \nmid n. \end{cases}$$

This defines a real linear functional on polynomials by finite sums. It introduces no measure or interchange of an infinite series with an integral. Its normalization and polynomial Stein identity are

$$L_{p,a}(1) = 1, \qquad L_{p,a}(xg) = pa\, L_{p,a}(g^{(p-1)}). \tag{27}$$

Indeed, on $g = x^n$, this is the recurrence $m_{n+1} = pa(n)_{p-1}m_{n-p+1}$, with the entire right-hand side defined as zero when $n < p - 1$. For $n + 1 = pj$, substitution of the displayed moments gives $(pj)!a^j/j!$ on both sides; when $p \nmid n + 1$, both sides vanish. Linearity proves the identity for every polynomial. Conversely this recurrence and $m_0 = 1$ determine each successive moment, hence the whole functional. Thus (27) characterizes this particular algebraic object uniquely.

Theorem 9.1. *For every $p \ge 2$ and real a,*

$$[L_{p,a}(g^2) \ge 0 \text{ for all } g \in \mathbb{R}[x]] \quad \iff \quad a = 0 \text{ or } (p = 2 \text{ and } a \ge 0).$$

Equivalently the functional is represented on all polynomials by a centered Gaussian variable, allowing variance zero. It is strictly positive on squares of nonzero polynomials exactly when $p = 2$ and $a > 0$.

Proof. If $a = 0$, the coefficient formula gives $L_{p,0}(g) = g(0)$, so square positivity holds. It is not strict since $g = x$ has zero square value. If $p = 2$ and $a \ge 0$, let Z be standard Gaussian and set $h = \sqrt{2a}$. All polynomials in hZ are integrable. Gaussian integration by parts gives

$$\mathbb{E}[hZg(hZ)] = h^2\mathbb{E}[g'(hZ)] = 2a\,\mathbb{E}[g'(hZ)].$$

The parts formula follows directly by integrating the derivative of $g(hz)e^{-z^2/2}$; polynomial growth makes the boundary terms vanish. Normalization and the just-proved uniqueness show $L_{2,a}(g) = \mathbb{E}[g(hZ)]$, which is square positive. When $a > 0$, its Gaussian density is strictly positive everywhere. A nonzero polynomial has finitely many zeros and is nonzero on some open interval, so the integral of its square is strictly positive. For $p = 2$, the necessary test $g = x$ gives $2a \geq 0$; strict positivity gives $2a > 0$.

It remains to exclude every $p > 2$ with $a \neq 0$, including odd orders and either sign of a. Put $b = p!a \neq 0$. Since p divides neither 2 nor $2(p-1)$, the moments give

$$L_{p,a}(x^2) = L_{p,a}((x^{p-1})^2) = 0, \qquad L_{p,a}(x\,x^{p-1}) = b.$$

For the second divisibility assertion, $p \mid 2p-2$ would imply $p \mid 2$, impossible when $p > 2$. Expanding the actual polynomial square now gives the explicit witness

$$L_{p,a}((x - bx^{p-1})^2) = -2b^2 < 0. \tag{28}$$

Thus no such functional is square positive or Gaussian represented. Conversely every Gaussian-represented functional is square positive, which proves the claimed equivalence. The preceding cases also settle strict positivity completely. □

The witness diagnoses an indefinite pairing before any completion. Both x and x^{p-1} have null square value, whereas their sum has square value $2b \neq 0$. Consequently the vectors with null square value do not even form an additive set. Quotienting those vectors cannot create a Hilbert space. Reversing the sign in (28) gives square value $2b^2 > 0$; the failure is indefiniteness rather than an identically zero pairing. These are exact polynomial identities, not conclusions about a conjectural positive path law or convergence of signed stochastic integrals.

The broader characteristic-function obstruction is classical: Lukacs [2, Theorem B, printed page 487] recalls Marcinkiewicz's theorem excluding an exponential of a polynomial of degree greater than two as a characteristic function. Our much narrower pure-order classification uses the two null diagonal moments directly and never assumes a characteristic-function representation of the formal exponential. It does not classify all degrees of the mixed $\mathcal{L}_{v,k}$ above, whose low-order variance is nonzero when $v > 0$. The strict finite truncation regions and their null compatibility boundaries remain the separate results of the preceding sections. No new Gaussian characterization or general Marcinkiewicz theorem is claimed.

10 Normalization, interpretation and exact formal scope

For $v > 0$, $x = \sqrt{v}\,y$ gives the normalized generating series $\exp(z^2/2 + \kappa z^3/6)$, $\kappa = k/v^{3/2}$. Reflection changes the sign of k; thus $r = \kappa^2$ governs the boundaries:

$$0 < r_4 < 1/5 < r_3 < 1/2 < 2.$$

Here r_4 solves $8 - 40r - 45r^3 = 0$ and $r_3 = (2\sqrt{7} - 4)/3$. Finite matching does not certify a positive all-degree functional; no all-degree classification of the mixed $\mathcal{L}_{v,k}$ is asserted. The separate pure-order theorem has its own hypotheses.

Gram matrices, Schur complements, finite spectral measures and null-polynomial compatibility are classical truncated Hamburger tools. Curto–Fialkow [1], Theorems 3.1/3.9, distinguish positivity from singular extendibility. The odd-case range condition appears concretely in (22) and (19); (18) gives the strict-case freedom. The bounded primary comparison has not established priority for our explicit parameter formulas or escape example.

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The general Jacobi argument, two-atom coordinates and quantitative total-variation/lower-tail bounds are labelled human proofs or deductions, distinct from the mapped named formal declarations. No new general moment theorem is claimed.

References

[1] R. E. Curto and L. A. Fialkow, Recursiveness, positivity, and truncated moment problems, *Houston J. Math.* **17** (1991), no. 4, 603–635. In particular Theorems 3.1 and 3.9. https://www.math.uh.edu/~hjm/v017n4/0603CURTO.pdf.

[2] E. Lukacs, Some extensions of a theorem of Marcinkiewicz, *Pacific Journal of Mathematics* **8** (1958), no. 3, 487–501. Theorem B in the introduction. https://msp.org/pjm/1958/8-3/pjm-v8-n3-p11-p.pdf.